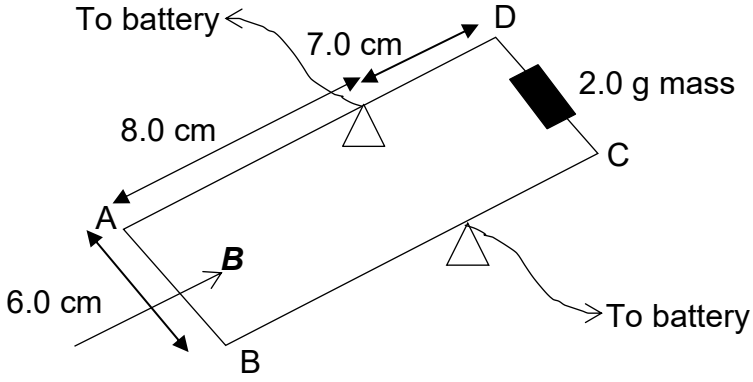
4. A light rectangular wire frame ABCD is used as a current balance. It has the following dimensions:

AB = CD = 6.0 cm
BC = DA = 15.0 cm

The wire frame is pivoted such that the pivot is 8.0 cm from AB, and 7.0 cm from CD. A 2.0 g mass is attached to CD. Side AB is entirely immersed in a uniform horizontal magnetic field ***B*** which is perpendicular to AB, as shown in the diagram below:



A current of 3.0 A flows through the wire frame, and as the result, the frame is found to come to a rest in the horizontal orientation.

(a) State the direction (AB or BA) of the current. [1]

(b) Calculate the magnetic field, ***B***. [3]

(c) Explain why such a current balance cannot be used to measure the strength of a magnetic field that is not horizontal. [3]